

DATA VISUALIZATION · FINAL INDIVIDUAL PROJECT

Global Video Game Sales

A story told through 10 analytical questions, 1980–2016

16,598 games analyzed

10 analytical questions

Interactive Streamlit dashboard



Project Overview

This project explores 37 years of the global video game industry (1980–2016) using a dataset of 16,598 titles with platform, genre, publisher, and regional sales data. Ten multi-dimensional analytical questions drive the story — each answered with a purpose-built Plotly visualization, then surfaced in a curated interactive dashboard.

Dataset

Kaggle — Video Game Sales (real-world, VGChartz-sourced). Numerical, categorical & temporal fields.

10 Questions

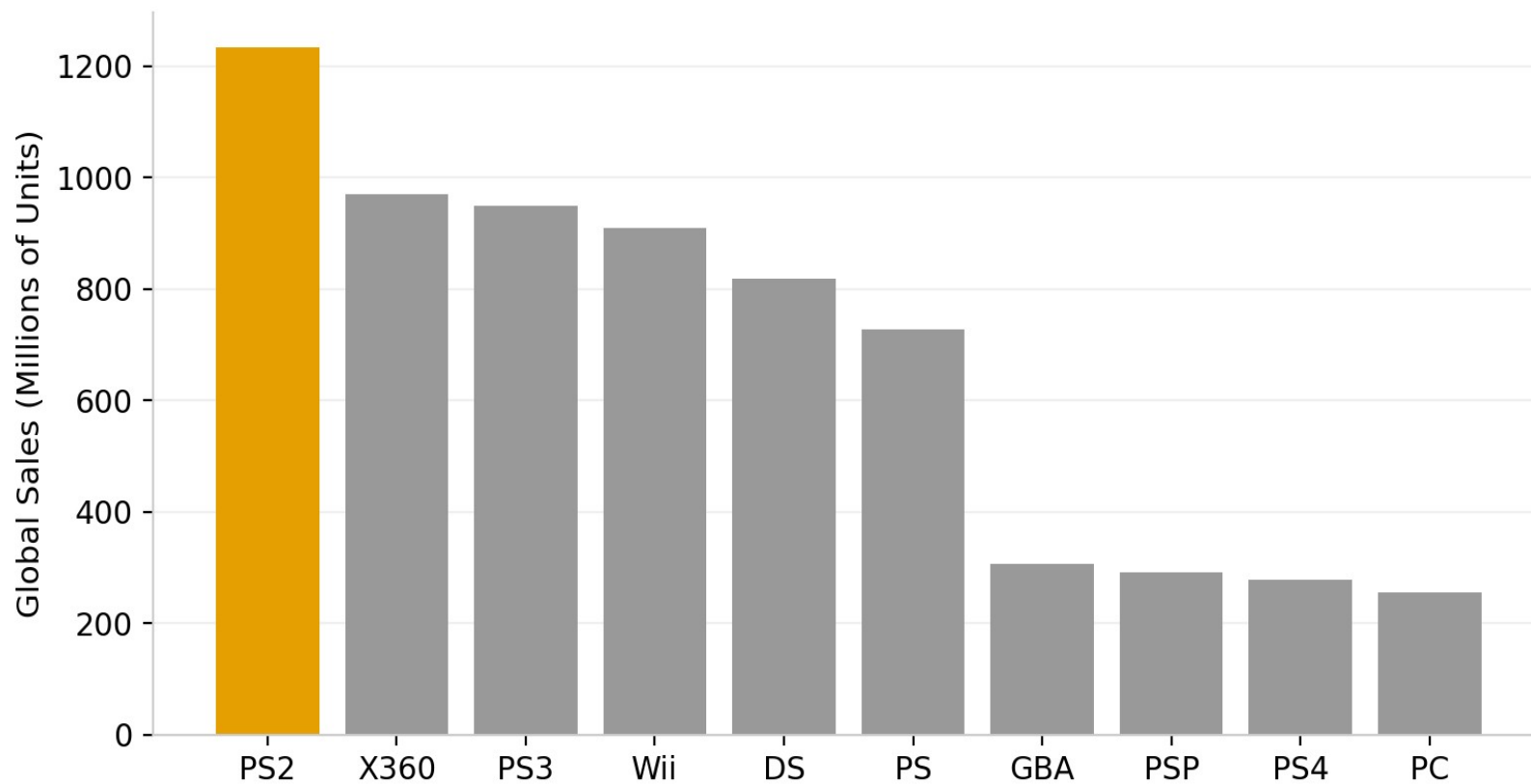
Comparisons across groups, trends over time, and cross-variable relationships — not simple counts.

Deliverables

Jupyter notebook (Plotly), this deck, and a live Streamlit dashboard on GitHub.

Q1 How does total sales vary across platforms?

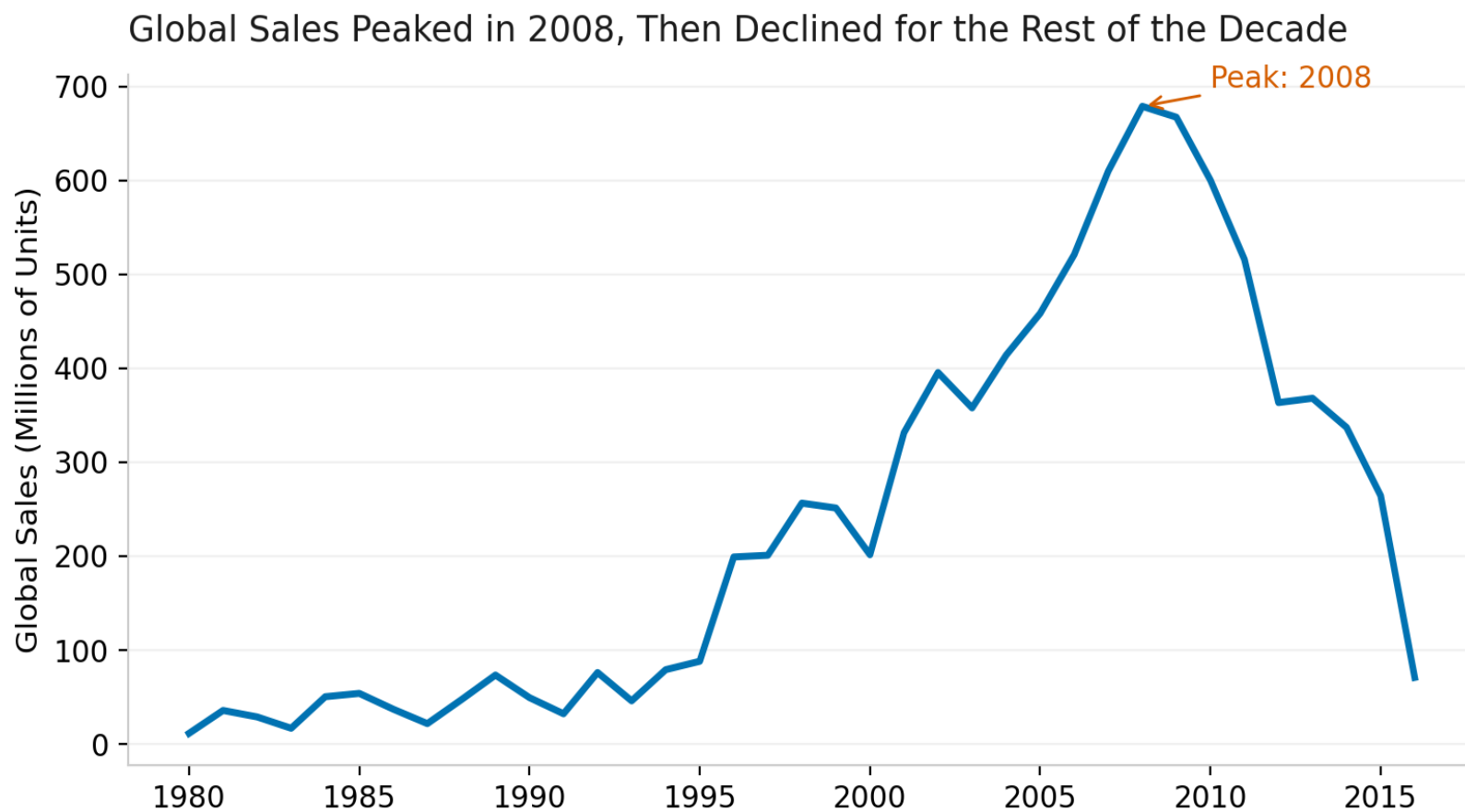
PS2 Outsold Every Other Platform, Nearly 20% Ahead of the X360



KEY INSIGHT

Sales are concentrated, not even. The PS2 alone accounts for roughly 1.2 billion units — far ahead of the X360 and PS3 — reflecting its unusually long, dominant console generation (2000–2013).

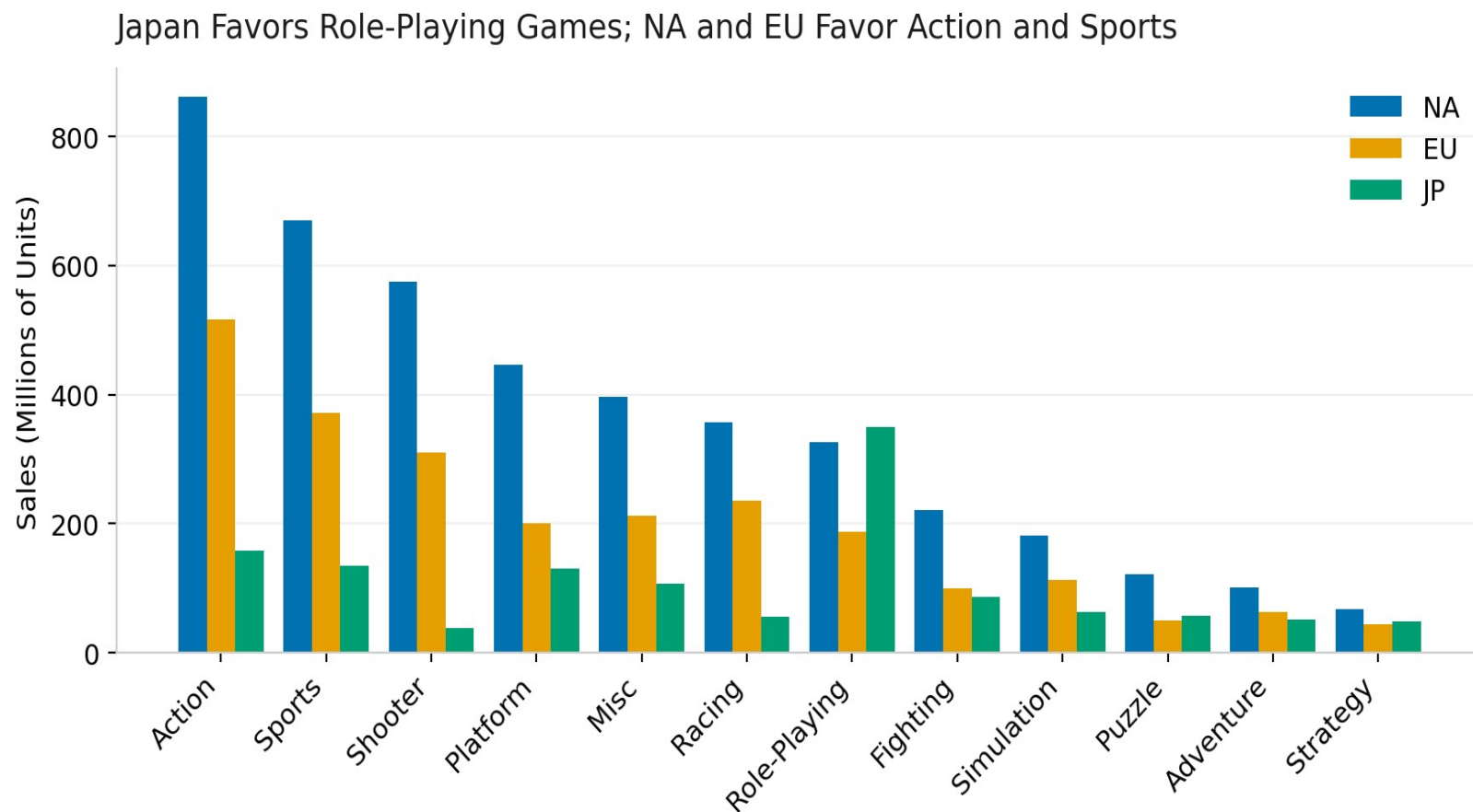
Q2 How have global sales trended over time?



KEY INSIGHT

Sales grew steadily through the 1990s and 2000s, peaked in 2008, then declined — coinciding with the rise of digital distribution, which this physical-sales dataset does not capture.

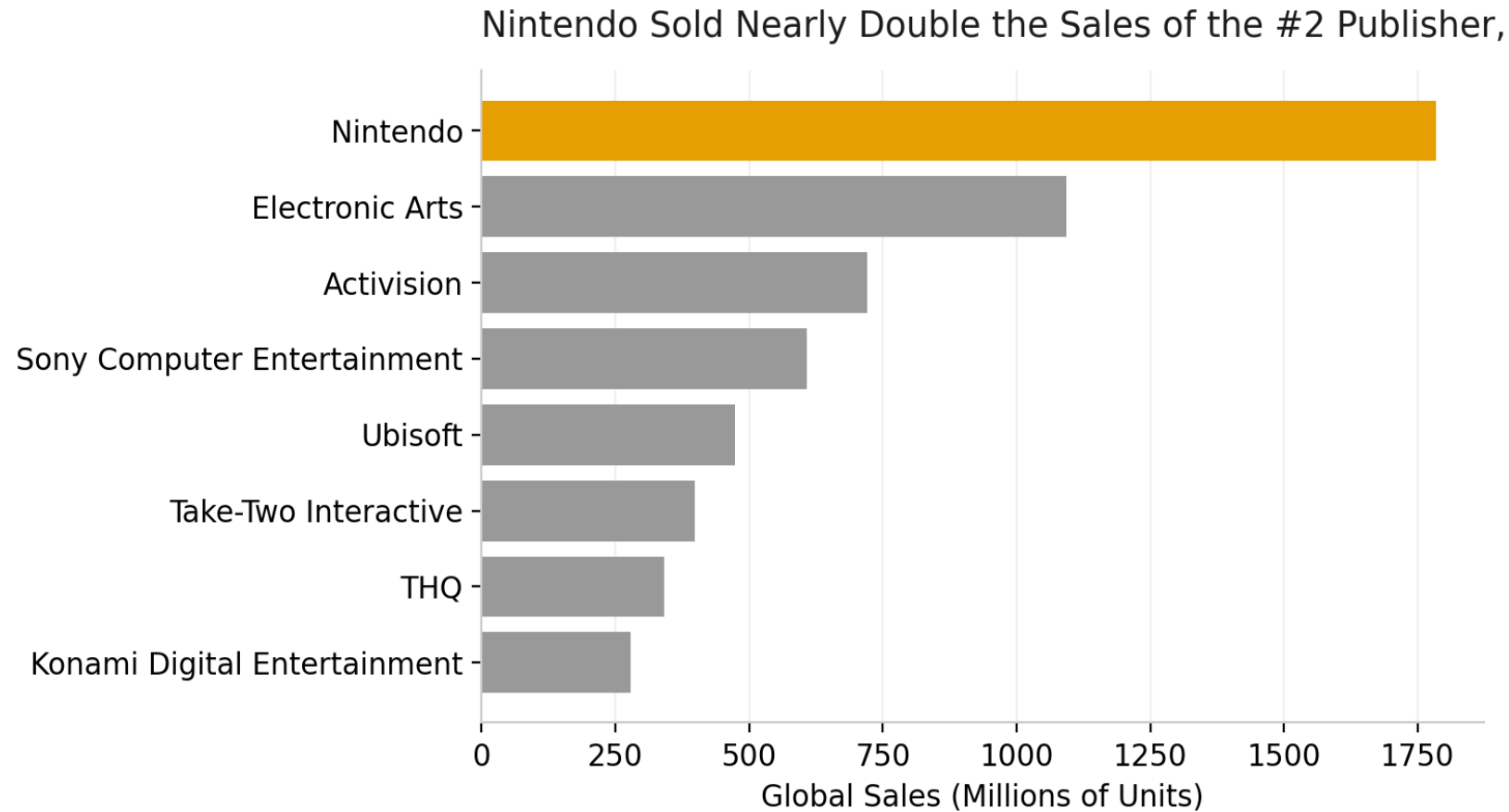
Q3 Do regional genre preferences differ?



KEY INSIGHT

NA and EU both peak on Action and Sports. Japan stands apart — its sales are proportionally concentrated in Role-Playing games, a clear cultural divergence in taste.

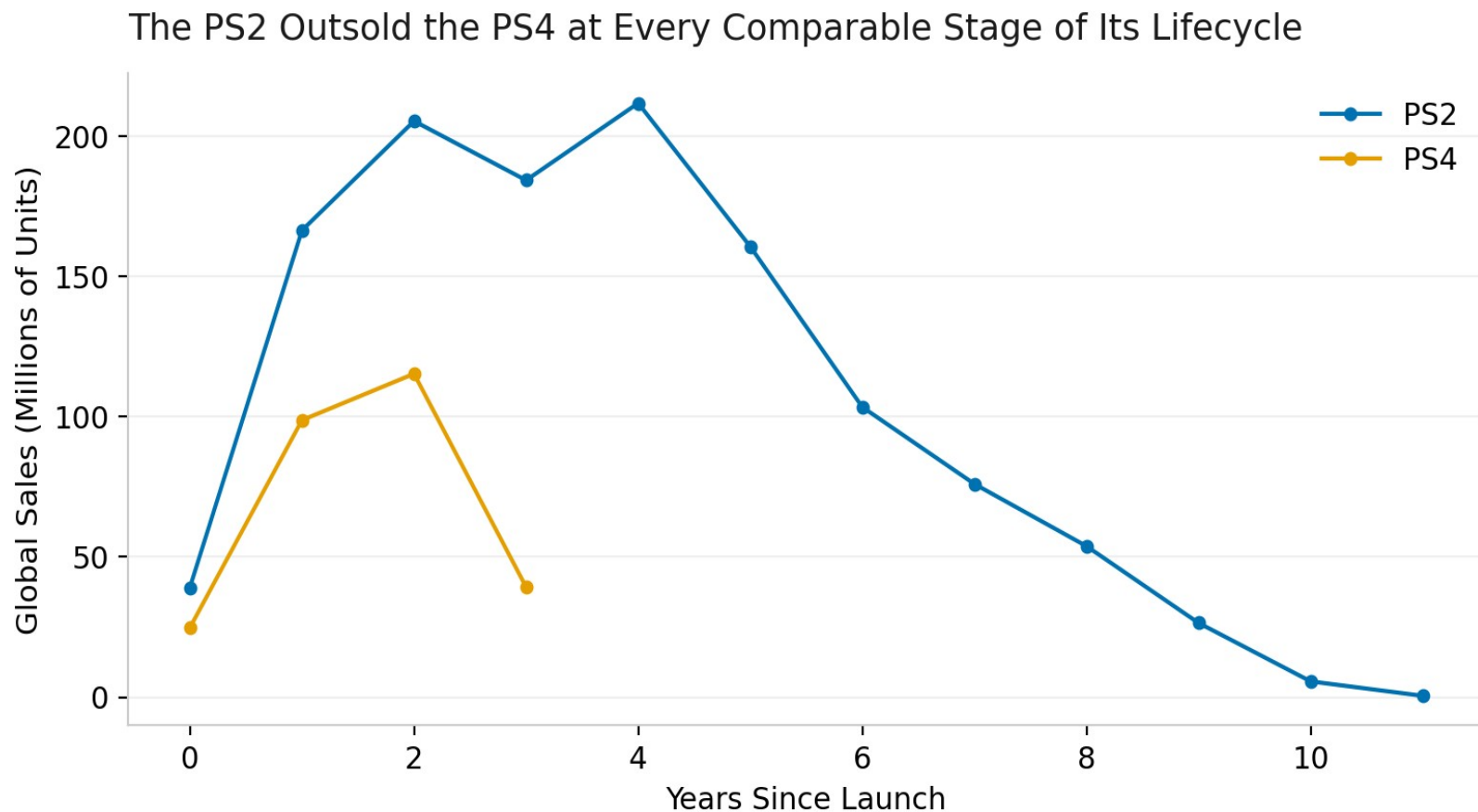
Q4 Which publishers dominate total sales?



KEY INSIGHT

Nintendo leads by a wide margin — nearly double Electronic Arts, the next closest — benefiting from owning both hardware and best-selling first-party franchises.

Q8 Do newer consoles outsell older ones at the same lifecycle stage?



KEY INSIGHT

In physical units, the PS2 outsold the PS4 at every comparable stage of its lifecycle. The industry's physical-sales peak hasn't been matched since — digital sales (not captured here) have taken over.



Key Findings

1 Sales are concentrated, not spread evenly — the PS2 alone accounts for ~1.2B units sold.

2 Global sales peaked in 2008, then declined — consistent with the industry's shift to digital distribution.

3 Regional taste diverges sharply: Japan over-indexes on RPGs; NA/EU favor Action and Sports.

4 Nintendo leads total publisher sales, but genre leadership is specialized across companies, not monopolized.

5 Console lifecycles follow a boom-then-decline curve — no modern platform has matched the PS2's lifecycle-adjusted peak.



Interactive Dashboard

A curated Streamlit dashboard surfaces the most important visuals with real interactivity: filters for year range, genre, and regional focus, organized across three tabs (Platforms & Publishers, Regional Trends, Sales Over Time). The same colorblind-safe palette and decluttered style used in the notebook carries through consistently.

GITHUB REPO

[paste your GitHub repo link here]

LIVE DASHBOARD (Streamlit Community Cloud)

[paste your deployed Streamlit URL here]

Snapshot: dashboard filters let the viewer explore sales by year range, genre, and region focus without leaving the browser.